Referee Report on Manuscript IP-105150.R1 Submitted to Inverse Problems

The authors have made revisions and provided a point-by-point response. While some presentational issues have been addressed, the following scientific concerns need to be further addressed before the manuscript can be accepted for publication.

1. **Main Concern: Insufficient Validation of the Adaptive Basis Function Selection in MA-RFM.**

   The central contribution of this paper is the MA-RFM framework, which improves source reconstruction by augmenting the approximation space with morphology-aware basis functions. This idea is innovative. The key challenge, however, lies in the automatic selection of these basis functions, and the authors have added Algorithm 3 in the revised manuscript to address this. Nevertheless, the current presentation raises some concerns.

   Algorithm 3 relies on geometric descriptors computed from an imprecise point cloud extracted from the coarse solution, combined with several manually chosen thresholds. The reliability of this procedure is inherently uncertain due to: (i) the imprecision of the coarse solution from which the point cloud is derived; (ii) the fact that computing discrete curvature requires an ordered boundary point sequence, yet the point cloud obtained by thresholding the gradient field is unordered — the paper provides no explanation of how this ordering is achieved; (iii) the absence of any theoretical justification or systematic sensitivity analysis for the chosen threshold values.

   More critically, in Examples 4.3–4.7, the basis functions used in MA-RFM are all directly specified as hyperparameter settings, with no indication that they were determined by Algorithm 3. From the presentation, it appears that the basis function types, the number of source regions, and the functional form of the source were all prescribed using prior knowledge of the true source, rather than being automatically inferred from the coarse solution. This significantly undermines the claim that MA-RFM can "adaptively identify critical regions and incorporate morphology-aware activation functions." The superiority of

MA-RFM under conditions where such prior knowledge is unavailable has not been demonstrated. While Table 9 in Appendix D.3 lists some geometric parameters and basis type classifications produced by Algorithm 3 for Examples 4.3–4.7, it is presented without any explanation and is not referenced anywhere in the main text. Furthermore, it is not connected to the experimental setups in Examples 4.3–4.7, making it impossible to verify whether Algorithm 3 was actually used to drive basis function selection in these experiments.

2. **Minor Issues.**

   (a) **Page 9, Line 54**: The summation index n does not appear in the summand. This is likely a typographical error.

   (b) **Page 18**: The stated center $(0.5, 0.5)$ of circle B is inconsistent with the values reported in Table 6 and Figure 11.

   (c) **Figures 3, 5, 7, 9, 11, 13, 15**: The term "piece-wise absolute error" appears in all error plot captions but is never defined in the manuscript. A precise mathematical definition should be provided at first use, for example, $e(x_i) = |S_M(x_i) - S_{ex}(x_i)|$ for $x_i \in P_{\text{test}}$.

   (d) **Figure 14 caption**: "The vertical black and horizontal red lines" should read "The horizontal black and vertical red lines."

   (e) **Table 9**: The radius estimate for Example 4.3 is reported as $r = 2.03 \times 10^{-2}$, whereas Table 3 reports $\hat{r} = 0.2034$ for the same example. These values differ by an order of magnitude. If this is a typographical error, it should be corrected. If it reflects a genuine difference in what is being reported, the authors should clarify what Table 9 actually represents and how it relates to the estimates used in the MA-RFM experiments.

   (f) **Structure**: Appendices C and D contain material that is central to understanding the proposed method. It is recommended that these appendices be incorporated into the main text at appropriate locations, in order to improve the coherence and readability of the manuscript.

Based on the above, I recommend major revision before the manuscript can be considered for publication. Specifically, the authors should first clarify, for each of Examples 4.3–4.7, whether the basis function types specified in the hyperparameter settings were determined by Algorithm 3 or prescribed using prior knowledge of the true source geometry. The authors should also provide at least one numerical experiment in which both the geometry and the functional form of the source are completely unknown, preferably non-trivial, with the source consisting of multiple disjoint components, and Algorithm 3 is demonstrably shown to drive the basis function selection end-to-end, with its outputs explicitly connected to the MA-RFM experimental setup. Furthermore, the implementation details of Algorithm 3 should be fully specified, including the DBSCAN parameters, the procedure for extracting an ordered boundary sequence from the unordered point cloud, and a justification for the classification thresholds t_{AR}, t_{CV}, t_{Σ}, t_p. The minor issues listed above should also be corrected.